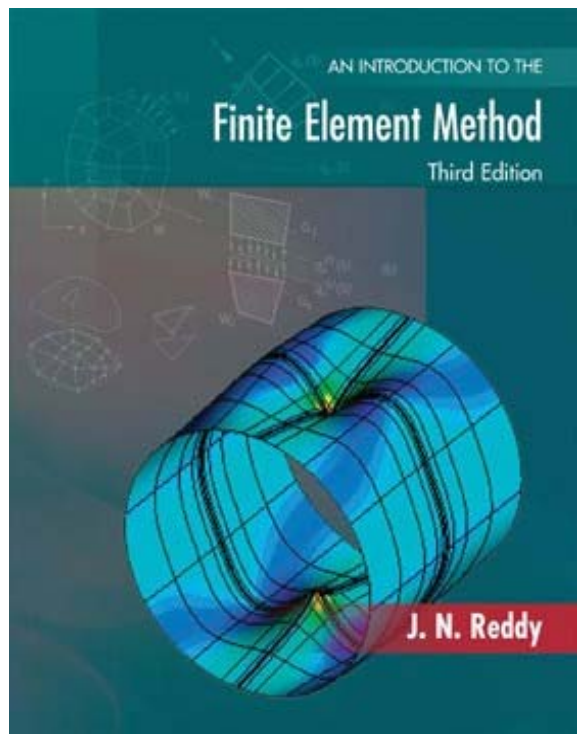


The Finite Element Method

Computer Program FEM2D

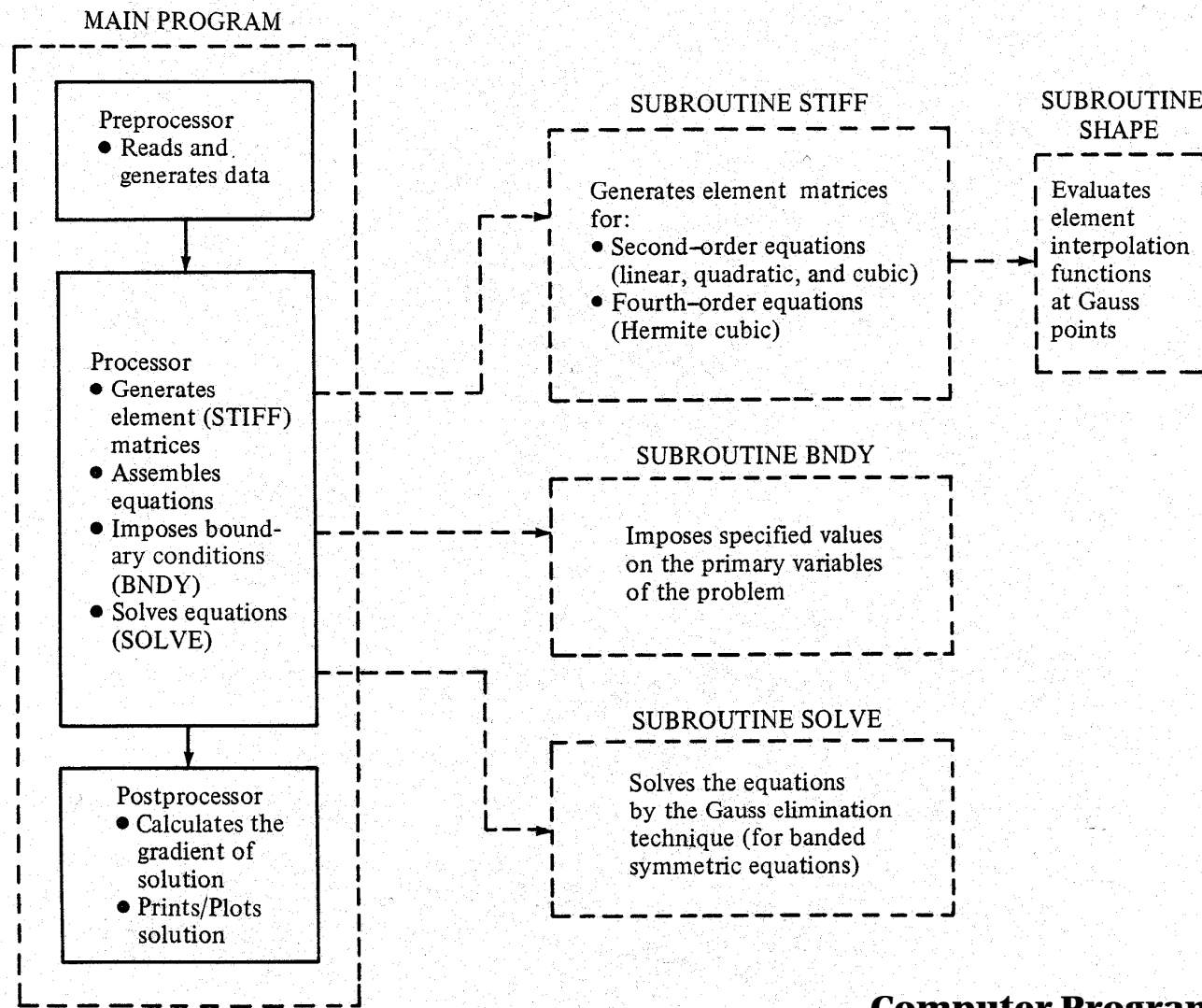
Read: Chapter 13



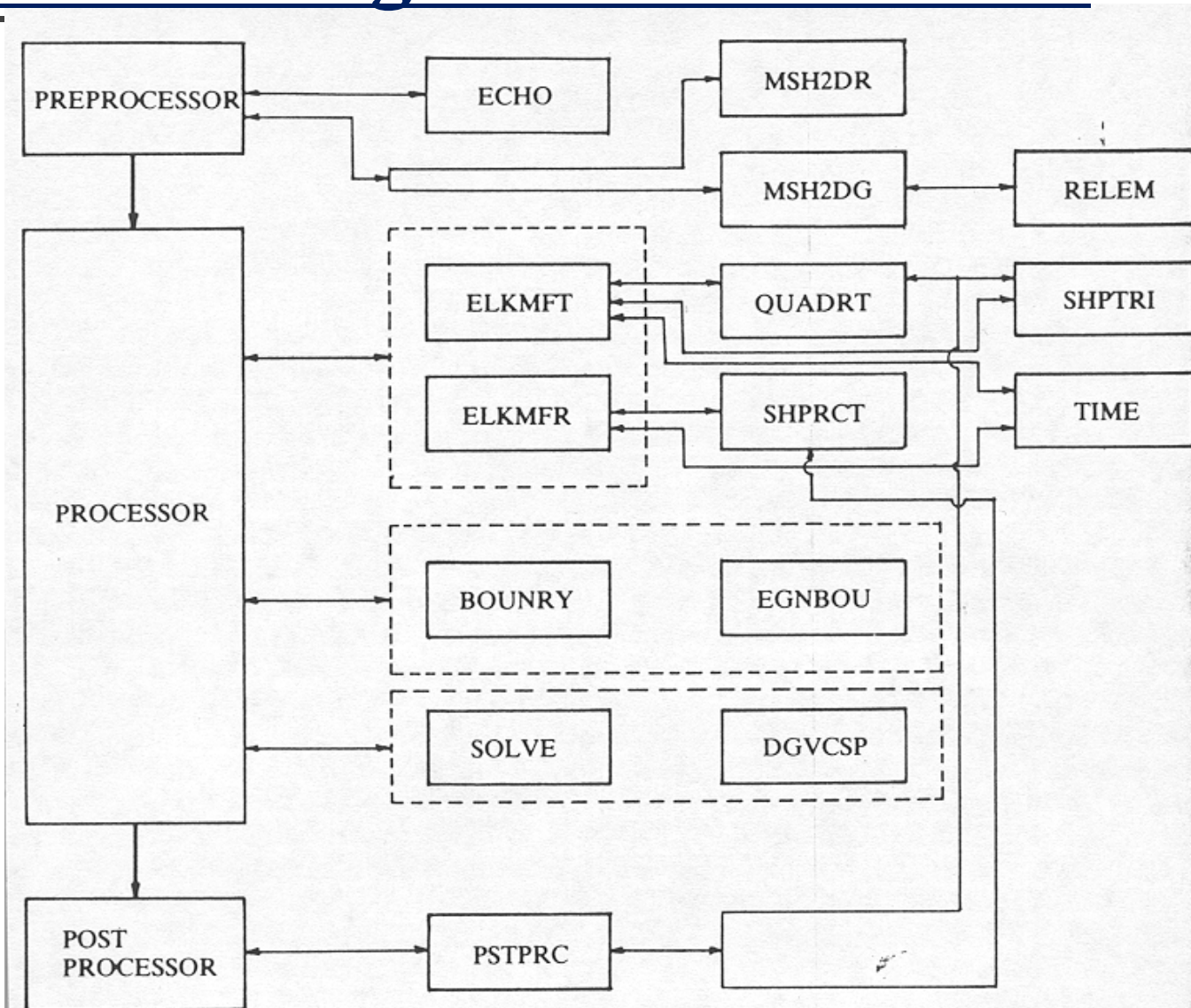
CONTENTS

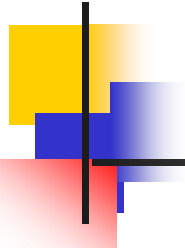
- Logical units of a FEM2D program
- Flow chart of a typical processor unit
- Element calculations
- Computer program FEM2D
- Input data to FEM2D
- Example problem for FEM2D

LOGICAL UNITS OF FEM2D



FEM2D Program FLOWCHART





FEM2D Program DATA INPUT

* Data Card 1:

TITLE - Title of the problem being solved (80 characters)

* Data Card 2:

ITYPE - Problem type

ITYPE = 0, Single variable problems

ITYPE = 1, Viscous incompressible flow problems

ITYPE = 2, Plane elasticity problems

ITYPE = 3, Plate bending problems by the FSDT

ITYPE = 4, Plate bending problems by the CPT(N)

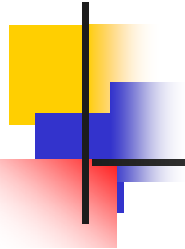
ITYPE = 5, Plate bending problems by the CPT(C)

IGRAD - Indicator for computing the gradient of the solution
or stresses in the postprocessor part of the program

IGRAD = 0, No postprocessing is required

IGRAD > 0, Postprocessing is required

When ITYPE=0 and IGRAD=1, the gradient is computed
as in (13.4.2) ; for ITYPE=0 and IGRAD > 1, the
gradient is computed as in Eq (13.4.3)



FEM2D Program DATA INPUT

ITEM - Indicator for dynamic analysis:

ITEM = 0, Static analysis is required

ITEM > 0, Either eigenvalue or transient analysis
is required:

ITEM = 1, Parabolic equation is analyzed

ITEM = 2, Hyperbolic equation is analyzed

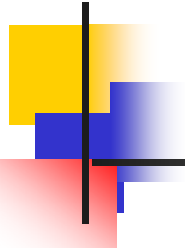
NEIGN - Indicator for eigenvalue analysis:

NEIGN = 0, Static or transient analysis is required

NEIGN > 0, Eigenvalue analysis is required:

NEIGN = 1, Frequency/vibration analysis

NEIGN > 1, Stability analysis of plates



FEM2D Program DATA INPUT

* Data Card 3: SKIP the card if NEIGN = 0

NVALU - Number of eigenvalues to be printed

NVCTR - Indicator for printing eigenvectors:

NVCTR = 0, Do not print eigenvectors

NVCTR > 0, Print eigenvectors

* Data Card 4:

IELTYP - Element type used in the analysis:

IELTYP = 0, Triangular elements

IELTYP > 0, Quadrilateral elements

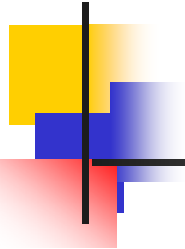
NPE - Nodes per element:

NPE = 3, Linear triangle (IELTYP=0)

NPE = 4, Linear quadrilateral (IELTYP>0)

NPE = 6, Quadratic triangle (IELTYP=0)

NPE = 8 or 9, Quadratic quadrilateral (IELTYP>0)



FEM2D Program DATA INPUT

MESH - Indicator for mesh generation by the program:

MESH = 0, Mesh is not generated by the program

MESH = 1, Mesh is generated by the program for rectangular domains (by MSH2DR)

MESH > 1, Mesh is generated by the program for general domains (by MSH2DG)

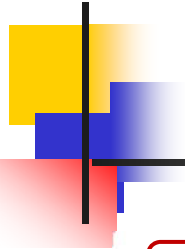
NPRNT - Indicator for printing certain output:

NPRNT = 0, Not print array NOD, element matrices, or global matrices

NPRNT = 1, Print array NOD and Element 1 matrices: [ELK] and {ELF}

NPRNT = 2, Print array NOD and assembled matrices, [GLK] and {GLF}

NPRNT > 2, Combination of NPRNT=1 and NPRNT=2



FEM2D Program DATA INPUT

**Cards 5-7 are read only
when you give the mesh
information**

*** Data Card 5:** SKIP the card if MESH.EQ.1

NEM - Number of elements in the mesh when the user inputs the mesh or the mesh is generated by MSH2DG

NNM - Number of nodes in the mesh when the user inputs the mesh or the mesh is generated by MSH2DG

*** Data Card 6:** SKIP the card if MESH.NE.0; otherwise, the card is read in a loop on the number of elements (N=1, NEM)

NOD(N,I)-Connectivity for the N-th element (I=1,NPE)

*** Data Card 7:** SKIP the card if MESH.NE.0

GLXY(I,J)-Global x and y coordinates of I-th global node in the mesh (J=1, x-coordinate, J=2, y-coordinate)
Loops on I and J are: ((J=1,2), I=1,NNM); the NNM pairs of (x,y)--coordinates are read sequentially



FEM2D Program DATA INPUT

***** SKIP Cards 8, 9, 10 and 11 unless MESH.GT.1 *****

**Cards 8-11 are
read only when
mesh is generated
for complicated
domains.**

*** Data Card 8:**

NRECL - Number of line records to be read in the mesh

*** Data Card 9:** Read the following variables NRECL times

NOD1 - First global node number of the line segment

NODL - Last global node number of the line segment

NODINC - Node increment on the line

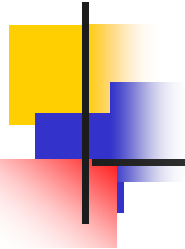
X1 - The global x-coordinate of NOD1

Y1 - The global y-coordinate of NOD1

XL - The global x-coordinate of NODL

YL - The global y-coordinate of NODL

RATIO - The ratio of the first element length to the last
element length



FEM2D Program DATA INPUT

* Data Card 10:

NRECEL - Number of rows of elements to be read in the mesh

* Data Card 11: Read the following variables NRECEL times

NEL1 - First element number of the row

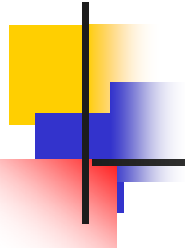
NELL - Last element number of the row

IELINC - Increment of element number in the row

NODINC - Increment of global node number in the row

NPE - Number of nodes in each element

NODE(I) - Connectivity array of the first element in the row
(I=1,NPE)



FEM2D Program DATA INPUT

* Data Card 12: SKIP the card if MESH.NE.1

NX - Number of elements in the x-direction

NY - Number of elements in the y-direction

Cards 12-14 are read only when mesh is generated for rectangular domains.

* Data Card 13: SKIP the card if MESH.NE.1

X0 - The x-coordinate of global node 1

DX(I) - The x-dimension of the I-th element (I=1,NX)

* Data Card 14: SKIP the card if MESH.NE.1

Y0 - The y-coordinate of global node 1

DY(I) - The y-dimension of the I-th element (I=1,NY)

* Data Card 15:

Cards 15 and 16 are read for all problems

NSPV - The number of specified primary variables



FEM2D Program DATA INPUT

* Data Card 16: SKIP the card if NSPV.EQ.0

ISPV(I,J) - Node number and LOCAL degree of freedom number of the I-th specified primary variable:

ISPV(I,1)=Node number; ISPV(I,2)=Local DOF number

The loops on I and J are: ((J=1,2),I=1,NSPV)

* Data Card 17: SKIP the card if NSPV.EQ.0 or NEIGN.NE.0

VSPV(I) - Specified value of the I-th primary variable
(I=1,NSPV)

**Cards 17-20 are NOT
read for eigenvalue
problems**

* Data Card 18: SKIP the card if NEIGN.NE.0

NSSV - Number of (nonzero) specified secondary variables

* Data Card 19: SKIP the card if NSSV.EQ.0 or NEIGN.NE.0

ISSV(I,J) - Node number and LOCAL degree of freedom number of the I-th specified secondary variable:

ISSV(I,1)=Node number; ISSV(I,2)=Local DOF number

The loops on I and J are: ((J=1,2),I=1,NSSV)



FEM2D Program DATA INPUT

* Data Card 20: SKIP the card if NSSV.EQ.0 or NEIGN.NE.0

VSSV(I) - Specified value of the I-th secondary variable
(I=1 NSSV)

Data Cards 21 - 27 are for SINGLE VARIABLE PROBLEMS (ITYPE=0) only

* Data Card 21: SKIP the card if ITYPE.NE.0

**Cards 21-23 are read
for all problems**

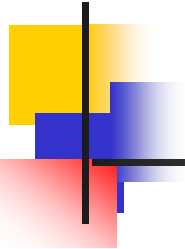
A10		
A1X		Coefficients of the differential equation
A1Y		$a_{11} = A10 + A1X * X + A1Y * Y$

* Data Card 22: SKIP the card if ITYPE.NE.0

A20		
A2X		Coefficients of the differential equation
A2Y		$a_{22} = A20 + A2X * X + A2Y * Y$

* Data Card 23: SKIP the card if ITYPE.NE.0

A00 - Coefficient of the differential equation



FEM2D Program DATA INPUT

* Data Card 23: SKIP the card if ITYPE.NE.0

A00 - Coefficient of the differential equation

* Data Card 24: SKIP the card if ITYPE.NE.0

ICONV - Indicator for convection boundary conditions
ICONV = 0, No convection boundary conditions
ICONV > 0, Convection boundary conditions present

* Data Card 25: SKIP the card if ITYPE.NE.0 or ICONV.EQ.0

NBE - Number elements with convection

* Data Card 26: SKIP the card if ITYPE.NE.0 or ICONV.EQ.0

The following cards are read for each I, I=1,NBE

IBN(I) - I-th element number with convection

INOD(I,J) - Array of the local node numbers of the side

BETA(I) - Film coefficient for convection on I-th element

TINF(I) - Ambient temperature of the I-th element

FEM2D Program DATA INPUT

* Data Card 28: SKIP the card if ITYPE.NE.1

AMU - Viscosity of the fluid

PENLTY - Penalty parameter used

**Card 28 is only read
for viscous flow problems
of Chapter 10**

____ Data Cards 29 & 30 are for PLANE ELASTICITY (ITYPE=2) only ____

* Data Card 29: SKIP the card if ITYPE.NE.2

LNSTRS - Flag for PLANE STRESS or PLANE STRAIN problems:

LNSTRS=0, Plane strain elastic problems

LNSTRS>0, Plane stress elastic problems

* Data Card 30: SKIP the card if ITYPE.NE.2

E1 - Young's moduli along the global x-axis

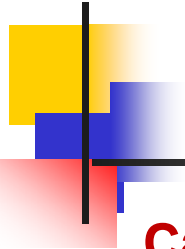
E2 - Young's moduli along the global y-axis

ANU12 - Poisson's ratio in the xy-plane

G12 - Shear modulus in the xy-plane

THKNS - Thickness of the plane elastic body analyzed

**Cards 29
and 30 are
read only
elasticity
problems
of Ch. 11**



FEM2D Program DATA INPUT

Card 31 is read only for plate bending problems of Chapter 12

*** Data Card 31:** SKIP the card unless ITYPE.GE.3

E1 - Young's moduli along the global x-axis

E2 - Young's moduli along the global y-axis

ANU12 - Poisson's ratio in the xy-plane

G12 - Shear modulus in the xy-plane

G13 - Shear modulus in the xz-plane

G23 - Shear modulus in the yz-plane

THKNS - Thickness of the plate analyzed

FEM2D Program DATA INPUT

* Data Card 32: SKIP the card if NEIGN.NE.0

Card 32 is read for all problems EXCEPT for eigenvalue problems

F0		
FX		Coefficients to define the source term:
FY		$f = F0 + FX*x + FY*y$

* Data Card 33: SKIP the card if ITEM.EQ.0

Card 33 is read for time-dependent and eigenvalue problems

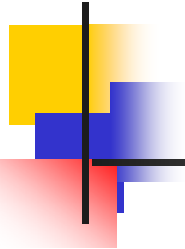
C0		
CX		Coefficients defining the temporal parts of the
CY		differential equations:

$CT = C0 + CX*x + CY*y$, when ITYPE = 0 or 1

$CT = (C0 + CX*x + CY*y)*THKNS$, when ITYPE = 2

$I0 = C0*THKNS$ |
 $I2 = C0*(THKNS**3)/12$ | when ITYPE=3 to 5
and CX and CY are not used | and NEIGN.LE.1

C0, CX, and CY denote the buckling parameters
when ITYPE.EQ.3 and NEIGN.GT.1



FEM2D Program DATA INPUT

* Data Card 34: SKIP the card if ITEM.EQ.0 or NEIGN.NE.0

Cards 34-37 are read for only time-dependent problems

NTIME - Number of time steps for the transient solution

NSTP - Time step number at which the source is removed

INTVL - Time step interval at which to print the solution

INTIAL - Indicator for nature of initial conditions:

INTIAL=0, Zero initial conditions are used

INTIAL>0, Non-zero initial conditions are used

* Data Card 35: SKIP the card if ITEM.EQ.0 or NEIGN.NE.0

DT - Time step used for the transient solution

ALFA - Parameter in the alfa-family of time approximation
used for parabolic equations:

ALFA=0, The forward difference scheme (C.S.)@

ALFA=0.5, The Crank-Nicolson scheme (stable)

ALFA=2/3, The Galerkin scheme (stable)

ALFA=1, The backward difference scheme (stable)



FEM2D Program DATA INPUT

GAMA - Parameter in the Newmark time integration scheme used for hyperbolic equations:

GAMA=0.5, Constant-average acceleration (stable)

GAMA=1/3, Linear acceleration scheme (C.S.)@

GAMA=0.0, The central difference scheme (C.S.)

ALFA = 0.5 for all schemes; For schemes for which
ALFA.LE.0.5 and GAMA < ALFA, DT is restricted to:
 $DT < 2/\text{SQRT}[\text{MAXEGN}*(\text{ALFA}-\text{GAMA})]$, MAXEGN being the
maximum eigenvalue of the discrete system.

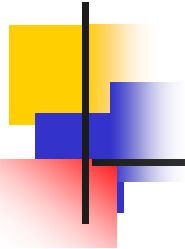
EPSLN - A small parameter to check if the solution has
reached a steady state

* Data Card 36: SKIP the card if ITEM or INTIAL.EQ.0, or NEIGN.NE.0

GLU(I) - Vector of initial values of the primary variables
(I=1,NEQ, NEQ=Number of nodal values in the mesh)

* Data Card 37: SKIP if ITEM.LE.1, NEIGN.NE.0, or INTIAL.EQ.0

GLV(I) - Vector of initial values of the first derivative
of the primary variables (velocity)
(I=1,NEQ, NEQ=Number of nodal values in the mesh)

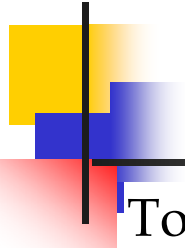


TO RUN THE EXECUTABLE PROGRAM **FEM2D.EXE**

Executable Computer Programs from the book, *An Introduction to the Finite Element Method* by J. N. Reddy, 3rd ed., McGraw–Hill, 2006.

Notes:

Programs FEM1D and FEM2D are a revised versions of the programs from the second edition of the book. The revisions are minor. Both FEM1D and FEM2D were compiled using the Microsoft Fortran compiler, with a fixed array dimensions. Hence, only a limited size problem can be analyzed using the compiled versions of the programs. The programs were compiled with a maximum number of degrees of freedom of 2,500. If your computer has the storage and you have the source programs, you may recompile the program after changing the DIMENSION statements in the programs.



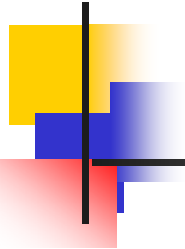
TO RUN THE EXECUTABLE PROGRAM **FEM2D.EXE**

To run a program on a PC, the files should be downloaded to your PC into a folder (say, *FEM_Reddy*). *The user is required to prepare a data file for each problem* he or she wants to solve, using the instructions in Table 7.3.2 for FEM1D and Table 13.4.1 for FEM2D. Most errors are mistakes made in the preparation of the data files. Therefore, you must check the data files when you see 'run-time error' message or the program is not executed (by returning just the echo of the input data file). All files should be in the same folder where FEM1D.EXE and FEM2D.EXE are placed.

To run the program:

- Double click on the executable file (say, FEM2D.EXE).
A window (called *Command Prompt* window) will pop open (with white letter and black background). It will read

File name missing or blank – please enter file name
UNIT 5?



TO RUN THE EXECUTABLE PROGRAM **FEM2D.EXE**

- Type the input file name (with its extension) and press *Enter*. For example, if the file name you prepared is labeled as **Prob1.inp**, you may type it as **prob1.inp** (not case sensitive). The Command Prompt window will now display

**File name missing or blank – please enter file name
UNIT 6?**

- Type the output file name (with its extension) and press *Enter*. For example, if you want the computer to return the output to file **Prob1.out**, you may type it as **prob1.out**. Note that you do not prepare this file; it will be created by the computer. The Command Prompt window will disappear, indicating that it has taken the data file and executed the program. You will find the file **prob1.out** in the same folder where you are running the program. Depending on the results you see, you may have to correct the data file.

Example 1: Example 8.3.1 of the book

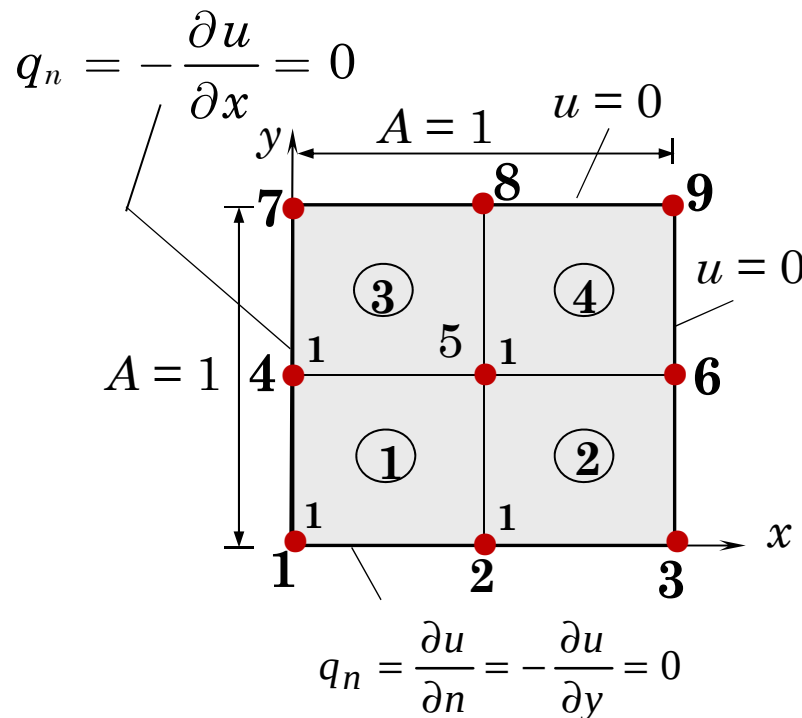
Problem description

Problem Type, ITYPE = 0;
steady-state, ITEM = 0; NEIGN = 0

$-\nabla^2 u = 1$ in a square of 2 units
($\alpha_{11} = \alpha_{22} = 1, \alpha_{00} = 0, f(x) = 1$)

Gradient computation is for heat transfer-like Problems, i.e., compute the x and y -components of the flux vector (IGRAD = 1)

Quadrant model



$$q_x = -a_{11} \frac{\partial u}{\partial x}, \quad q_y = -a_{22} \frac{\partial u}{\partial y}$$

FEM Mesh and boundary conditions

2 x 2 mesh of linear rectangular elements
NX = NY = 2; IELTYP = 1; NPE = 4; MESH = 1
X0 = Y0 = 0.0; DX(1) = 0.5, DX(2) = 0.5
DY(1) = 0.5, DY(2) = 0.5
NSPV = 5, Global nodes 3, 6, 7, 8, and 9
have specified PVs with specified values being
zeros; no convection



Example 1: Example 8.3.1 of the book

INPUT DATA to FEM2D

Ex 8.3.1: Solution of the Poisson equation on a square domain (quadrant)

0	1	0	0						ITYPE,IGRAD,ITEM,NEIGN	
1	4	1	0						IELTYP,NPE,MESH,NPRNT	
2	2								NX,NY	
0.0	0.5	0.5							X0,DX(I)	
0.0	0.5	0.5							Y0,DY(I)	
5									NSPV	
3	1	6	1	7	1	8	1	9	1	ISPV(I,J)
0.0	0.0	0.0	0.0	0.0	0.0					VSPV(I)
0										NSSV
1.0	0.0	0.0								A10, A1X, A1Y
1.0	0.0	0.0								A20, A2X, A2Y
0.0										A00
0										ICONV
1.0	0.0	0.0								F0, FX, FY

Example 2: Find the eigenvalues of the problem in Example 1

$$\frac{\partial u}{\partial t} - \nabla^2 u = 0$$

$$(c = 1, a_{11} = a_{22} = 1, a_{00} = 0)$$

Problem description

Problem Type, ITYPE = 0; IGRAD = 0

Eigenvalue analysis, ITEM = 1; NEIGN = 1

FEM Mesh and boundary conditions

4 x 4 mesh of linear rectangular elements

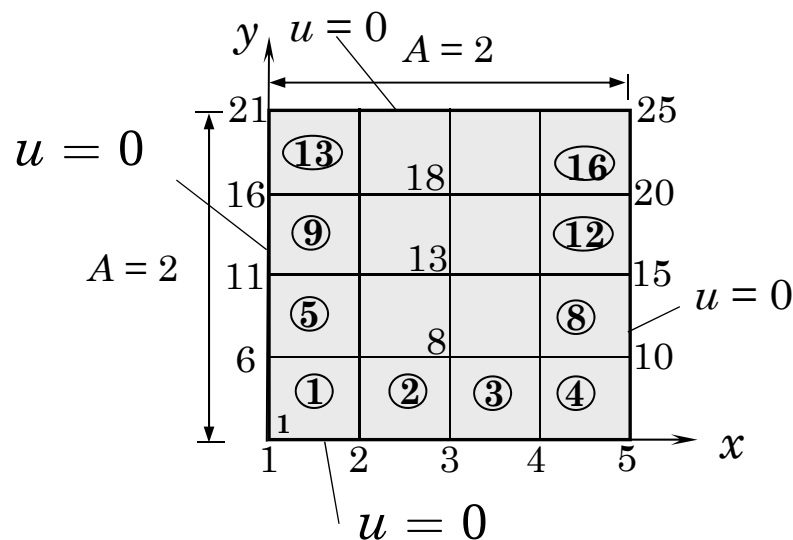
NX = NY = 4; IELTYP = 1; NPE = 4; MESH = 1

X0 = Y0 = 0.0; DX(1) = 0.5, DX(2) = 0.5,

DX(3) = 0.5, DX(4) = 0.5, DY(1) = 0.5,

DY(2) = 0.5, DY(3) = 0.5, DX(4) = 0.5

NSPV = 16, Global nodes on the boundary have specified homogeneous PVs.



Example 2: INPUT DATA to FEM2D

Example 2: Eigenvalues of the Poisson equation on a square domain (full)

0 1 1 1	ITYPE,IGRAD,ITEM,NEIGN
9 1	NVALU, NVCTR
1 4 1 0	IELTYP,NPE,MESH,NPRNT
4 4	NX,NY
0.0 0.5 0.5 0.5 0.5	X0,DX(I)
0.0 0.5 0.5 0.5 0.5	Y0,DY(I)
16	NSPV
1 1 2 1 2 1 4 1 5 1	
6 1 10 1 11 1 15 1 16 1	
20 1 21 1 22 1 23 1 24 1 25 1	ISPV(I,J)
1.0 0.0 0.0	A10, A1X, A1Y
1.0 0.0 0.0	A20, A2X, A2Y
0.0	A00
0	ICONV
1.0 0.0 0.0	C0, CX, CY

Example 3: Find the transient response of the problem in Example 1

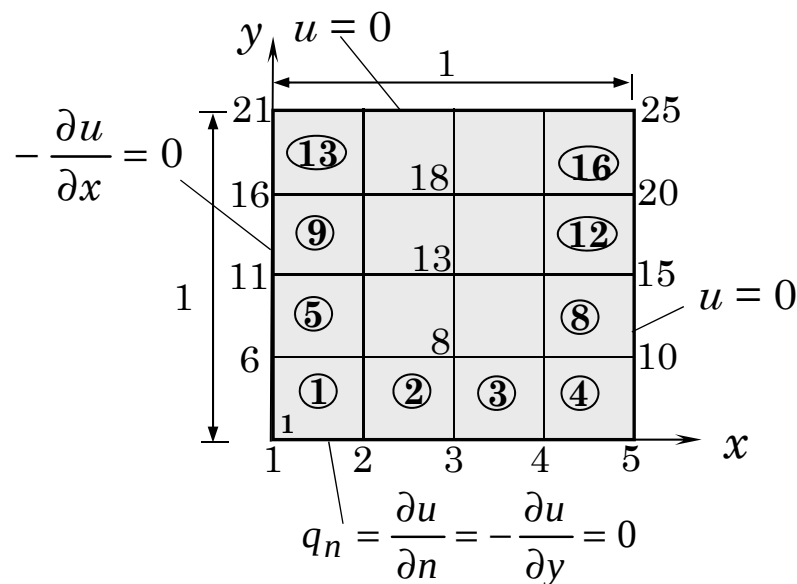
Problem description

$$\frac{\partial u}{\partial t} - \nabla^2 u = 0$$

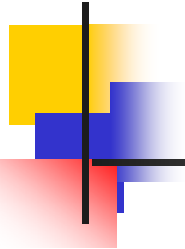
$$(c = 1, a_{11} = a_{22} = 1, a_{00} = 0)$$

Problem Type, ITYPE = 0; IGRAD = 0
Transient analysis, ITEM = 1; NEIGN = 1

FEM Mesh and boundary conditions



4 x 4 mesh of linear rectangular elements
NX = NY = 4; IELTYP = 1; NPE = 4; MESH = 1
X0 = Y0 = 0.0; DX(1) = 0.25, DX(2) = 0.25,
DX(3) = 0.25, DX(4) = 0.25, DY(1) = 0.25,
DY(2) = 0.25, DY(3) = 0.25, DX(4) = 0.25
NSPV = 16, Global nodes on the boundary
have specified homogeneous PVs.



Example 3: INPUT DATA to FEM2D

Example 8.6.3: Transient analysis of an parabolic equation (membrane)

0	0	1	0							ITYPE,IGRAD,ITEM,NEIGN								
1	4	1	0							IELTYP,NPE,MESH,NPRNT								
4	4							NX,NY										
0.0	0.25	0.25	0.25	0.25						X0,DX(I)								
0.0	0.25	0.25	0.25	0.25						Y0,DY(I)								
9							NSPV											
5	1	10	1	15	1	20	1	21	1	22	1	23	1	24	1	25	1	ISPV
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	VSPV
0							NSSV											
1.0	0.0	0.0							A10, A1X, A1Y									
1.0	0.0	0.0							A20, A2X, A2Y									
0.0							A00											
0							ICONV											
0.0	0.0	0.0							F0, FX, FY									
1.0	0.0	0.0							C0, CX, CY									



Example 3: INPUT DATA to FEM2D (continued)

100	101	1	1		
0.025	0.5	0.5	1.0E-3		
				NTIME,NSTP,INTVL,INTIAL	
				DT,ALFA,GAMA,EPSLN	
0.400	0.375	0.300	0.175	0.0	
0.375	0.35156	0.28125	0.16406	0.0	
0.300	0.28125	0.225	0.13125	0.0	
0.175	0.16406	0.13125	0.076563	0.0	
0.0	0.0	0.0	0.0	0.0	Initial cond., GLU(I)
0.0	0.0	0.0	0.0	0.0	
0.0	0.0	0.0	0.0	0.0	
0.0	0.0	0.0	0.0	0.0	
0.0	0.0	0.0	0.0	0.0	
0.0	0.0	0.0	0.0	0.0	Initial cond., GLV(I)

Example 8.6.4: Transient analysis of a rectangular membrane (hyperbolic)

0	0	2	0											ITYPE,IGRAD,ITEM,NEIGN				
1	4	1	0											IELTYP,NPE,MESH,NPRNT				
4	4											NX,NY						
0.0	0.5	0.5	0.5	0.5											X0,DX(I)			
0.0	0.25	0.25	0.25	0.25											Y0,DY(I)			
9											NSPV							
5	1	10	1	15	1	20	1	21	1	22	1	23	1	24	1	25	1	ISPV
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	VSPV
0											NSSV							
12.5	0.0	0.0											A10, A1X, A1Y					
12.5	0.0	0.0											A20, A2X, A2Y					
0.0											A00							
0											ICONV							
0.0	0.0	0.0											F0, FX, FY					
2.5	0.0	0.0											C0, CX, CY					



Example 4: INPUT DATA to FEM2D (continued)

100	101	1	1		
0.025	0.5	0.5	1.0E-3		
				NTIME,NSTP,INTVL,INTIAL	
				DT,ALFA,GAMA,EPSLN	
0.400	0.375	0.300	0.175	0.0	
0.375	0.35156	0.28125	0.16406	0.0	
0.300	0.28125	0.225	0.13125	0.0	
0.175	0.16406	0.13125	0.076563	0.0	
0.0	0.0	0.0	0.0	0.0	Initial cond., GLU(I)
0.0	0.0	0.0	0.0	0.0	
0.0	0.0	0.0	0.0	0.0	
0.0	0.0	0.0	0.0	0.0	
0.0	0.0	0.0	0.0	0.0	
0.0	0.0	0.0	0.0	0.0	Initial cond., GLV(I)